

ARIAN SALARI

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EDUCATION

University of California, Irvine

B.S. in Mechanical Engineering, Honors Collegium

Irvine, CA

Expected June 2028

Irvine Valley College

Associate's in Engineering, Honors Program, Dean's List, Cumulative GPA: 3.95

Irvine, CA

August 2024 – May 2026

ENGINEERING EXPERIENCE

Rezvani Motors

Dec 2024 – Mar 2025

Mechanical Engineering Intern

Irvine, CA

- Designed SolidWorks CAD models and technical drawings for vehicle components, incorporating physical measurements and calculations to support design revisions and prototype development.
- Observed CFD-based aerodynamic evaluations that assess airflow behavior and support vehicle design decisions.
- Supported prototype development and manufacturing by reviewing drawings, inspecting components for fitment, and communicating design observations with engineering and production teams.

PROJECTS & RESEARCH

Autonomous Solar-Assisted Quadcopter UAV, Personal Project

May 2026 – Present

- Designed a parametric quadcopter airframe in SolidWorks and 3D-printed custom structural components for fitment testing, assembly, and iterative design refinement.
- Built a solar-powered autonomous charging station for a ~74 Wh battery system, enabling self-recharge between flights without manual battery replacement.
- Integrating ~20 W of lightweight onboard photovoltaics with autonomous return-to-dock functionality to reduce battery power demand and extend operational autonomy.

Aircraft Winglet Aerodynamics and Efficiency, IVC / Saddleback Research Symposium

Aug 2025 – Jan 2026

- Designed and 3D-printed baseline, sharklet, and split-scimitar wingtip configurations for comparative aerodynamic testing in physical and digital wind-tunnel environments.
- Quantified relative wingtip-vortex strength using wake-string deflection and image-based displacement measurements, with the split-scimitar reducing average deflection from 1.53 in to 0.98 in, a ~36% reduction versus the baseline.
- Extended prior award-winning research that earned Second Place in Fluids and Aerodynamics at the Orange County Science and Engineering Fair.

Medical Device Validation Research, IVC / Saddleback Research Symposium

Aug 2024 – Jan 2025

- Investigated Normothermic Ex-Vivo Heart Perfusion (NEHP) as a physiological testing platform for artificial heart valve validation under controlled flow and pressure conditions.
- Compared FDA-approved mechanical validation methods with industry testing practices, identifying limitations in replicating physiological conditions and evaluating NEHP-perfused porcine hearts as a higher-fidelity testing approach.

Shape-Memory Alloy Robotic Hand, OC Science and Engineering Fair

Nov 2023 – Mar 2024

- Designed and built a five-finger robotic hand actuated by Nitinol shape-memory alloy wire, using electrical current to induce contraction and drive finger articulation.
- Refined the actuation mechanism across multiple prototypes, using elastic return elements to restore finger position and improve coordinated motion across all five fingers.

LEADERSHIP

Irvine Valley College STEM Club

Mar 2025 – May 2026

President

Irvine, CA

- Led hands-on engineering projects and competition teams, including rocket-design builds, while organizing technical workshops, industry speaker events, and research/internship opportunities.

Irvine Valley College Computer Science Club

May 2025 – May 2026

Vice President

Irvine, CA

- Organized programming workshops, technical speaker sessions, and hackathon preparation while connecting members with internships and research opportunities.

SKILLS & INTERESTS

Technical Skills: Mechanical Design, GD&T, Technical Drawings, Design Iteration, Fitment Testing, FDM & Resin 3D Printing, Lean Six Sigma Principles, 5S, Visual Management, Scrum Master Certification (International Scrum Institute)

Software: SolidWorks, SimFlow, AutoCAD, Fusion 360, Excel, GitHub, Visual Studio

Programming: MATLAB, Python, C++, Java

Interests: Traveling, World History, Airplane Spotting, Flying, Guitar, Philosophy, Camping, Martial Arts, Culinary Arts